

One Model, Many Geometries: Spectral Dynamics of Multilingual LLMs

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1 Literature Review

We introduce works relevant to ours. First, we survey existing approaches to the study of multilingual training dynamics; then, we introduce macro-level spectral metrics as a complementary approach. Finally, we survey existing open-source multilingual LLMs, identifying suitable ones for our study.

Multilingual Training Dynamics. Several works explore the emergence of multilingual capabilities in LLMs using feature-level approaches. [Riemenschneider and Frank \(2025\)](#) and [Wang et al. \(2024\)](#) both trace neuron development over pretraining: the former finds that cross-lingual abstractions emerge over time, while the latter shows high correlation between neuron overlap and zero-shot cross-lingual transfer. [Bayazit et al. \(2026\)](#) takes a more mechanistic approach, training cross-coders across checkpoints to detect feature emergence and discontinuation. [Blevins et al. \(2022\)](#) show that cross-lingual transfer in encoder-only XLM-R is asymmetric and acquired at different pretraining stages depending on language.

Macro-Level Representation Geometry. Unlike feature-level approaches, macro-level spectral methods capture the global organization of the representation space. [Garrido et al. \(2023\)](#) show these properties correlate with task accuracy, and their use is motivated by the anisotropy of LLMs’ representation spaces ([Ethayarajh, 2019](#)). [Li et al. \(2025\)](#)—the direct motivation for our work—applies them to the study of pretraining dynamics. They use RankMe ([Garrido et al., 2023](#))—a spectral metric of effective rank ([Roy and Vetterli, 2007](#))—to identify three geometry phases during LLM pretraining: initial compression, entropy-seeking expansion, and final compression. However, their analysis is limited to English models and data.

Open-Source Multilingual LLMs. Analyzing multilingual training dynamics requires open-

source models with public pretraining checkpoints. Prior work has used BLOOM ([Workshop et al., 2023](#)) (560M–7B parameters, 43 languages), but only 8 checkpoints are available. Models with more checkpoints include Lucie ([Gouvert et al., 2025](#)) (40 checkpoints) and the Tri Series¹ (14 checkpoints), though both support fewer than 6 languages. We identify Apertus ([Apertus et al., 2025](#)) (71 checkpoints, 15T tokens, 1800+ languages) and FuxiTranyu ([Sun et al., 2024](#)) (58 checkpoints, 600B tokens, 40+ languages) as the most suitable models: both are available in an 8B size, and their differing checkpoint frequencies and token counts make them complementary subjects of study.

2 Contributions of Our Work

Our work differs from previous ones by (1) applying macro-level geometry analysis to multilingual training dynamics, (2) providing a more granular analysis, and (3) connecting findings to downstream differences in performance. Concretely, we make the following contributions:

- 1. Multilingual representation geometry:** we extend the work of [Li et al. \(2025\)](#) to the multilingual setting, complementing existing feature-level approaches.
- 2. Fine-grained training dynamics:** leveraging Apertus-8B and FuxiTranyu-8B, we conduct a more granular analysis of multilingual training dynamics than prior work.
- 3. Geometry-to-performance connection:** rather than studying capabilities acquired during distinct pretraining phases, we investigate whether *differences* in geometry evolution correlate with *differences* in downstream monolingual and cross-lingual performance.

¹<https://huggingface.co/collections/trillionlabs/tri-series>

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